

Closure properties of CFLs.

Union. Suppose A and B are CFLs where $L(G_1) = A$, $L(G_2) = B$ and the start symbols are S_1 for G_1 and S_2 for G_2 .

Construct a grammar G s.t. $L(G) = A \cup B$ as follows:

Ensure that G_1 and G_2 have disjoint set
[rename the nonterminals if required]

Combine the productions of G_1 & G_2 .

Add a new start symbol S and the productions: $S \rightarrow S_1$, $S \rightarrow S_2$

Concatenation. if A and B are CFLs with $L(G_1) = A$ and $L(G_2) = B$ construct G s.t.
 $L(G) = AB = \{xy \mid x \in A, y \in B\}$ as follows:

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Concatenation if A and B are CFLs with $L(G_1) = A$ and $L(G_2) = B$ with start symbols S_1 & S_2 . Construct G such that $L(G) = AB = \{xy \mid x \in A, y \in B\}$ as follows:

- Combine the grammars G_1 & G_2 .
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Kleene star. if A is a CFL with $L(G_1) = A$ and start symbol S_1 , construct G s.t. $L(G) = A^*$ as follows:

- Take G_1 along with a new start symbol S along with the production
$$S \rightarrow S_1 S \mid \epsilon.$$

Intersection. CFLs are not closed under intersection.

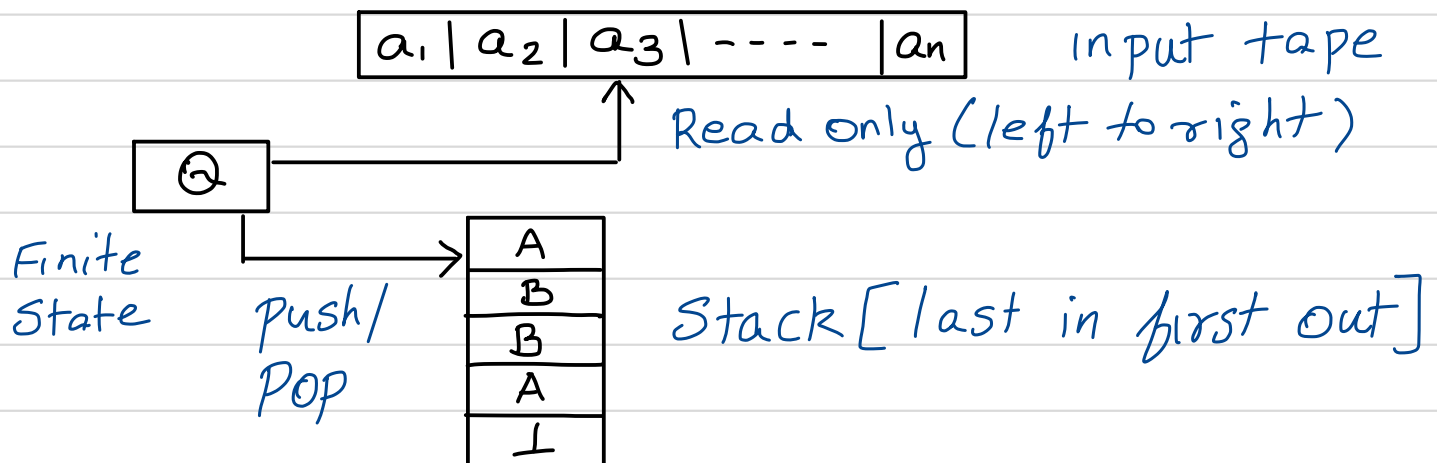
$$\{a^m b^m c^n \mid m, n \geq 0\} \cap \{a^m b^n c^n \mid m, n \geq 0\}$$

$$= \underbrace{\{a^n b^n c^n \mid n \geq 0\}}$$

Not a CFL.

Non deterministic pushdown automata (NPDA)

Finite automata + a stack.



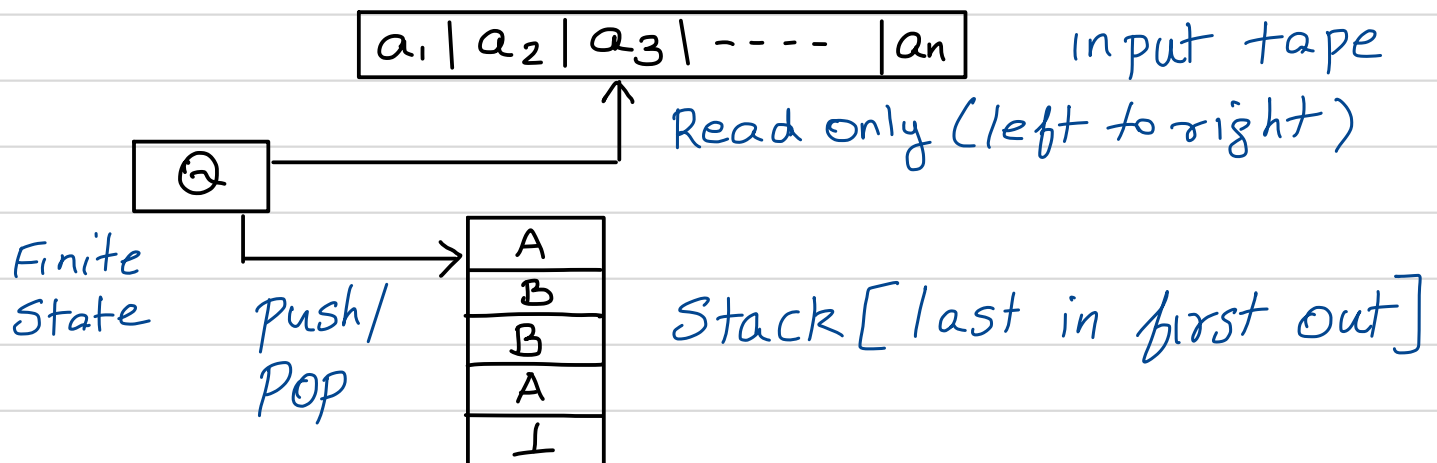
Working of the machine

- pops the top symbol off the stack.
- Makes a transition based on the top of the stack, input symbol and current state.

Transition: push a sequence of symbols onto the stack, change state, move the read head one cell to the right

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ϵ -transitions are allowed: Machine can pop and push without reading an input symbol or moving the input head pointer.

Stack can store unbounded information but access is limited

Definition of a nondeterministic PDA.

$$M = (Q, \Sigma, \Gamma, s, \delta, \perp, F)$$

Q - finite set of states, Σ - finite set: input alphabet

$s \in Q$ - start state, $F \subseteq Q$: set of final / accept States.

Γ - finite set: Stack alphabet

$\perp$ - initial stack symbol.

$$S \subseteq (Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma) \times (Q \times \Gamma^*)$$

$$((p, a, A), (q, B_1, \dots, B_k)) \in S : \text{Example 1}$$

$\downarrow \quad \quad \downarrow$
last First

$$((p, \epsilon, A), (q, B_1, \dots, B_k)) \in S : \text{Example 2}$$

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Configurations.: A configuration of M is

an element of $Q \times \Sigma^* \times \Gamma^*$.

$\downarrow$ current state $\downarrow$ current contents of the stack
 $\rightarrow$ part of the input that is unread.

Start Configuration: $(s, \epsilon, \perp)$

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Start Configuration: $(\delta, \epsilon, \perp)$

Define: 1 step next configuration relation $\xrightarrow{1}_M$

if $((p, a, A), (q, \gamma)) \in \delta$ then for any $y \in \Sigma^*$ and $\beta \in \Gamma^*$

$$(p, ay, A\beta) \xrightarrow{1}_M (q, y, \gamma\beta)$$

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Let $\xrightarrow{*}_M$ denote the reflexive transitive closure of $\xrightarrow{1}_M$

Acceptance. Two types: By final state and empty stack.

By final state: M accepts x by final state if

$$(q, x, \perp) \xrightarrow{*}_M (q, \epsilon, \gamma) \text{ for some } q \in F, \gamma \in \Gamma^*$$

$\hookrightarrow$ can be any string.

By empty stack. M accepts x by empty stack if

$$(q, x, \perp) \xrightarrow{*}_M (q, \epsilon, \epsilon) \text{ for some } q \in Q.$$

$\hookrightarrow$ can be any state

$L(M)$ - set of all strings $x \in \Sigma^*$ accepted by M .

Example: Set of balanced parentheses.

NPDA - accepting by empty stack.

$$Q = \{q\}, \Sigma = \{[, \cdot]\}, \Gamma = \{\perp, [\cdot]\} \quad \delta = q.$$

$$((q, \perp, \perp), (q, [\cdot])) \in \delta.$$

$$((q, \perp, [\cdot]), (q, [[\cdot])) \in \delta$$

$$((q, \cdot, [\cdot]), (q, \epsilon)) \in \delta.$$

$$((q, \epsilon, \perp), (q, \epsilon)) \in \delta.$$

NPDA : a few observations.

- $\perp$ is treated as any other stack symbol.
Useful to define the start configuration
Need not always stay at the bottom of the stack.
- A transition $((p, a, A), (q, B))$ does not apply unless
 A is on the top of the stack.
if the stack is empty then no transition applies
- To accept by empty stack, everything must be
popped and nothing pushed back.
The stack must be empty in a configuration.